

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination May 2017

PH212 Physical Pharmaceutics

Date: 15/05/2017, Monday

Time: 01:30 p.m. to 04:30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any TWO of the following.
- A Classify various viscometers. Describe two viscometer with diagram to find out viscosity of Non-Newtonian fluids. **10**
- B Describe the various factors affecting viscosity of solution. Write in detailed note on Non-Newtonian systems. **10**
- C What is Buffer capacity? How it is calculated? Write a note on Isotonic solutions with respect to measurements and methods of adjusting tonicity. **10**
- Q 2** Attempt any TWO of the following.
- A Describe the terms polymorphs and liquid crystal with suitable example. How it is characterized? Explain. **10**
- B What is spreading Coefficient? Derive its equation. Describe the HLB scale and method of calculation of HLB by different techniques. **10**
- C (a) Briefly describe the methods for measurement of surface and interfacial tensions. **05**
- (b) Write a note on surface active agents. Discuss the electric properties of interfaces. **05**

SECTION- II

- Q 3** Attempt any TWO of the following.
- A Enlist various methods for determination of particle size and explain sieving method and optical microscopy in detail. **10**
- B (a) Describe the derived properties of powder. **05**
- (b) Describe the methods for determining surface area. **05**
- C Explain various type of Emulsion. Write a note on physical stability of emulsion. **10**
- Q 4** Attempt any TWO of the following.
- A Differentiate Flocculated and deflocculated Suspension. What is meant by controlled flocculation? How controlled flocculation can be achieved? **10**
- B Classify the pharmaceutical complexes. Give the methods of preparation and analysis of inclusion complex. **10**
- C (a) Explain protective colloidal action in detail. **05**
- (b) Describe the types and properties of colloids. **05**